

Generation Interconnection Request System Impact Study Report

For Queue Project Q12-0169

Watermill 100 kV
August 23, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0169. The Interconnection Customer proposes a 137.2 MW wind facility interconnecting at the Watermill 100 kV substation in the Ohio area. The request was received on October 31, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 114 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$8,085,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Newport Tie / Post Rock 230kV: 110.4% loading (667 MVA) following the outage of the Baver / Trego Substation 500kV line; DFAX = 11.8%.

For the loss of Trego / Beaver 500kV, the Weedpatch / Canal Rd 230kV loads to 107.9% of its 394 MVA rating. Project DFAX: 9.6%.

The Myrtle Point / Post Rock 230kV is overloaded to 106.5% of its 756 MVA emergency rating for the loss of the Trego / Beaver 500kV Line. The project contributes 9.0% DFAX (12.4 MW) to this overload.

The Orchard Road / Ft Humbug 500kV is overloaded to 101.7% of its 2,551 MVA emergency rating for the loss of the Ft Humbug / W Campus 345kV. The project contributes 13.4% DFAX (18.4 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Newport Tie / Post Rock 230kV	Rebuild line with higher-capacity conductor	\$25,661,000	14% (\$3,545,000)
Weedpatch / Canal Rd 230kV	Reconductor line	\$8,488,000	18% (\$1,549,000)
Post Rock Substation / Myrtle Pt 230kV Line	Reconductor line	\$10,719,000	11% (\$1,220,000)
Orchard Road / Ft Humbug 500kV	Reconductor line	\$41,743,000	4% (\$1,771,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
	Total allocated to project		\$8,085,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.